6.6 Thermal characteristics

The operating junction temperature T_J must never exceed the maximum given in *Table 23: General operating conditions*.

The maximum junction temperature in °C that the device can reach if respecting the operating conditions, is:

$$T_J(max) = T_A(max) + P_D(max) \times \Theta_{JA}$$

where:

- $T_A(max)$ is the maximum operating ambient temperature in °C,
- Θ_{JA} is the package junction-to-ambient thermal resistance, in °C/W,
- $P_D = P_{INT} + P_{I/O}$,
  - P_{INT} is power dissipation contribution from product of I_{DD} and V_{DD}
  - $P_{I/O}$ is power dissipation contribution from output ports where $P_{I/O} = \Sigma (V_{OL} \times I_{OL}) + \Sigma ((V_{DDIO1} - V_{OH}) \times I_{OH})$, taking into account the actual V_{OL} / I_{OL} and V_{OH} / I_{OH} of the I/Os at low and high level in the application.

Table 70. Thermal resistance

Symbol	Parameter	Package(1)	Value	Unit
Θ_{JA}	Thermal resistance junction-ambient	UFQFPN20	76.4	°C/W
		TSSOP20	88.7	
		WLCSP12	148	
		SO8N	100	
Θ_{JB}	Thermal resistance junction-board	UFQFPN20	30	°C/W
		TSSOP20	54.6	
		WLCSP12	116.3	
		SO8N	56	
Θ_{JC}	Thermal resistance junction-case	UFQFPN20	31	°C/W
		TSSOP20	25.9	
		WLCSP12	10.6	
		SO8N	46	

1. Refer to *Section 6: Package information* for package dimensions

6.6.1 Reference documents

[1] *Integrated Circuits Thermal Test Method Environmental Conditions - Natural Convection (Still Air)* (JESD51-2A), JEDEC, January 2008. Available from www.jedec.org.